

From Rogue Employees to Rogue Agents: Repurposing Insider Threat Detection for AI Agent Governance

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Abstract—AI agent monitoring and insider threat detection share the same architecture: profile a baseline, flag deviations, encode assumptions about trust. We make this concrete across thirteen experiments. Using a Unified Behavioural Feature Schema (UBFS) that maps both employee activity logs and agent execution traces into a shared representation, we apply three anomaly detection models—Isolation Forest, LSTM Autoencoder, and Deep Clustering—across five domains. Cross-domain transfer is real but must be read carefully. An Isolation Forest trained on 329,000 insider-threat user-days scores 0.71 AUC-ROC on agent traces versus 0.73 in-domain—a 0.02 gap that falls within the seed-to-seed standard deviation (± 0.047), so we frame this as transfer within noise rather than 97% “retention.” This apparent robustness is partly an evaluation artifact: TRAIL is $\sim 97\%$ error-positive, so AUC-ROC rests on only a handful of negatives and AUC-PR (≈ 0.98) largely tracks the base rate; tellingly, a detector trained on TRAIL itself scores lower (0.58) than one transferred from CERT (0.71), which no genuine skill transfer would produce. We therefore report AUC-PR and recall at fixed low false-positive rates alongside AUC-ROC throughout. But the blind spots transfer too. Model distillation attacks span a detection spectrum from trivially caught (FOCUSED extraction: 0.99 AUC-ROC) to near-chance (HYDRA distributed: 0.54), and task decomposition—splitting a malicious objective into innocent subtasks—costs 5–6% detection power for IF and DC ($p=0.031$, Wilcoxon signed-rank), while LSTM Autoencoder shows near-invariance (-0.7% , $p=0.094$). Synthetic OWASP profiling identifies Tool Misuse (ASI02) as a blind spot (~ 0.52 AUC-ROC), but real-data validation on 500 ATBench trajectories inverts that ranking exactly: ASI02 is the best-detected of six categories at 0.82–0.95, and Spearman ρ between synthetic and real per-category rankings is -1.000 for all three models. The cause is visible in the generator—the ASI02 profile perturbs only privilege features while preserving behavioural structure, so it is undetectable by construction—and we report the divergence itself as a validity finding: synthetic perturbation profiling cannot locate blind spots, only restate the assumptions used to build them. Augmenting UBFS with eight semantic features (UBFS-28) improves detection by up to 13% on real data. Distillation sensitivity analysis shows detection degrades monotonically with attack intensity for FOCUSED/BROAD/COT profiles, while HYDRA remains near-chance regardless of scale. Adversarial evasion testing reveals that mimicry attacks degrade detection by up to 25%, with Memory & Context Poisoning (ASI06) most vulnerable. Temporal window ablation finds that 10 spans are optimal for Isolation Forest on both TRAIL and ATBench. Mapping five MITRE ATLAS techniques [1] into the UBFS confirms that the framework generalises beyond OWASP: all five techniques are strongly detectable (>0.93

AUC-ROC for IF/LSTM), introducing no new blind spots. The detection models port across domains. So do their biases, and so do their blind spots.

Index Terms—AI agent monitoring, insider threat detection, anomaly detection, cross-domain transfer, behavioural feature schema, AI governance, OWASP, MITRE ATLAS, adversarial evasion, model distillation

I. Introduction

An employee badges into a secure facility at 2 AM, copies files to a USB drive, and leaves. An AI agent invokes a privileged API at 2 AM, exfiltrates training data through a tool call, and terminates. The first is an insider threat. The second is a rogue agent. The monitoring systems that would detect them are architecturally identical.

Both systems profile a baseline of “normal” behaviour—one for employees, one for agents. Both flag deviations from that baseline as anomalous. Both encode assumptions about trust, authority, and what counts as legitimate action. Yet the two research communities have developed largely in isolation: the security community builds User and Entity Behaviour Analytics (UEBA) systems [2]; the AI safety community builds agent monitoring frameworks from scratch [3].

This separation is costly. The security community has spent two decades learning which behavioural features predict insider threats, which attack patterns evade detection, and which modelling assumptions introduce blind spots. The AI safety community is learning these same lessons from scratch—different conferences, different datasets, different terminology for the same underlying problem. If the structural analogy holds, this duplication is unnecessary. Worse, it means the agent monitoring community will rediscover the same blind spots the security community has already documented.

This paper tests whether the structural analogy is more than a metaphor. We take three anomaly detection models validated on the CMU-CERT insider threat dataset [4], apply them to AI agent execution traces through a unified feature schema, and measure what transfers, what breaks, and what biases come along for the ride.

The results are stronger than expected in one direction, and more concerning than expected in another. Cross-

domain transfer holds up—an insider threat model is not degraded on agent traces (a change within seed noise), and, on a balanced tool-calling benchmark where the metric is trustworthy, transfer to MCP traces measurably exceeds within-domain performance. But some blind spots transfer too—and telling those apart from the ones we invented turned out to be the harder problem. Distributed extraction is invisible to per-account monitoring because coordination is a property of the joint distribution and a per-entity detector only sees marginals; Anthropic’s February 2026 disclosure of campaigns spread across 20,000+ accounts is independent corroboration [5]. Tool Misuse (ASI02) looked like a second instance of the same thing, and by the standard synthetic protocol it is: an agent using the right tools in the right order with escalated privileges, structurally undetectable for the same reason insider “Scenario 2” attacks evade detection in [4]. On real data it is the most detectable category we measure. The synthetic result was a restatement of the generator’s own definition, and nothing inside the experiment could have shown that.

A. Research Questions

- RQ1: Can insider threat detection models detect anomalous agent behaviour through a shared behavioural feature schema?
- RQ2: Which OWASP Agentic Security Index categories are detectable by structural anomaly detection, and which are blind spots?
- RQ3: What governance assumptions embedded in insider threat models transfer to the agent monitoring domain?
- RQ4: Do emerging real-world attacks exploit the predicted blind spots, and can semantic feature augmentation close them?

B. Contributions

- 1) UBFS: A 20-dimensional Unified Behavioural Feature Schema mapping both insider threat indicators and agent trace signals into a shared representation across 7 categories.
- 2) Transfer evidence: Empirical demonstration that an Isolation Forest trained on 329,000 insider user-days is not degraded on agent traces (AUC-ROC: 0.731 $\rightarrow$ 0.711, a change within seed noise ± 0.047), with the caveat that TRAIL’s $\sim 97\%$ positive prevalence makes its AUC-ROC an unstable estimand (Section IV-B).
- 3) OWASP detection matrix: A per-category analysis under the standard synthetic protocol, showing three detection tiers—strong (ASI06: 0.975), moderate (Excessive Agency, ASI10: 0.70–0.86), and apparent blind spot (ASI02: 0.51–0.52). Contribution 9 shows why the third tier should not be believed.

- 4) Governance audit: Identification of six normality assumptions that transfer across domains, with analysis of their implications for monitoring fairness.
- 5) Distillation detection spectrum: Four distillation sub-profiles spanning AUC-ROC 0.54 (HYDRA distributed) to 0.99 (FOCUSED extraction), revealing a coordination blind spot: per-account behaviour in a distributed cluster is within normal bounds because coordination lives in the joint distribution a per-entity detector never observes.
- 6) Decomposition evasion: Quantified evidence that splitting malicious objectives into innocent sub-tasks costs 5–6% detection power for IF and DC ($p=0.031$), while LSTM shows near-invariance (-0.7% , $p=0.094$), mirroring insider threat “boiling frog” escalation.
- 7) MCP transfer generalisation: Cross-domain transfer validated on ATBench tool-calling traces, where TRAIL $\rightarrow$ ATBench Isolation Forest achieves 104.8% of within-domain performance.
- 8) UBFS-28 hybrid: Semantic augmentation improves four categories on synthetic data and adds +0.4 to +17.2% on real data, with the largest gains on the weakest categories. It leaves synthetic ASI02 at ~ 0.56 , which contribution 9 shows is uninformative rather than a negative result.
- 9) A validity finding for synthetic OWASP profiling: On 500 ATBench trajectories the synthetic per-category ranking is exactly inverted ($\rho = -1.000$, all three models), with the identified blind spot ASI02 turning out to be the best-detected category (0.52 synthetic vs. 0.82–0.95 real). The mechanism is that perturbation profiles measure their own definitions. We know of no way to detect this failure from inside the synthetic protocol, which makes real-data validation a precondition for blind-spot claims rather than a robustness check on them.
- 10) Distillation sensitivity: Feature-space scaling across 4 intensity levels shows monotonic detection degradation for FOCUSED/BROAD/COT profiles, while HYDRA remains near-chance (~ 0.52) at original intensity regardless of scale.
- 11) Adversarial evasion framework: Three evasion strategies—feature clamping, gradual escalation, and mimicry—quantifying the robustness of UBFS-based detection. Mimicry is the most effective, degrading ASI06 detection by up to 25%.
- 12) Temporal window transfer: Window ablation showing 10 spans optimal for Isolation Forest on both TRAIL and ATBench, paralleling Paper 1’s 7-day optimal window—a structural insight about the primacy of early warning signals.
- 13) MITRE ATLAS mapping: Five ATLAS adversarial ML techniques mapped to UBFS perturbation profiles, all expressible in the schema and all landing in the “strong” tier (>0.93 AUC-ROC for IF/LSTM).

Being a synthetic mapping, this establishes expressiveness, not detectability.

II. Related Work

Insider threat detection has evolved from rule-based systems through supervised classification to behavioural analytics [2], [6]. Each generation hits the same bottleneck: labeled data. Unsupervised methods sidestep this by modelling normal behaviour and flagging deviations [7]. Isolation Forest [8] exploits the geometric property that outliers require fewer random splits to isolate. LSTM autoencoders [9] learn temporal dependencies in behavioural sequences. Deep clustering [10] combines representation learning with cluster structure. Tuor et al. [11] applied deep learning to structured cybersecurity streams; CapPELLI et al. [12] established the practitioner framework for insider risk.

AI agent monitoring is a younger field. The OWASP Top 10 for Agentic Applications [3] taxonomises risks from prompt injection to excessive agency. The TRAIL dataset [13] provides annotated agent execution traces with error labels. The TRACE dataset [14] captures reward hacking in iterative self-refinement—agents that produce correct outputs through illegitimate means. Pan et al. [15] formalised reward misspecification as a safety concern.

Real-world AI agent attacks. In February 2026, Anthropic disclosed two categories of attack against Claude [5]: model distillation (three Chinese AI labs created $\sim 24,000$ fraudulent accounts and conducted 16M+ exchanges to extract capabilities) and AI-orchestrated espionage (state-sponsored group GTG-1002 used Claude Code with MCP tools as an autonomous hacking agent). The distillation attacks employed three distinct strategies—focused extraction, multi-capability probing, and chain-of-thought elicitation—plus a “hydra cluster” architecture distributing traffic across 20,000+ accounts. The ATBench benchmark [16] provides multi-turn, tool-centric safety scenarios with fine-grained risk categorisation, enabling evaluation on MCP-style traces. The AgentHarm dataset [17] provides 440 malicious agent tasks in both direct and decomposed configurations, enabling comparison of single-turn attacks against multi-step task decomposition—the same evasion strategy used in the GTG-1002 campaign.

Adversarial ML threat taxonomies. MITRE ATLAS [1] extends the ATT&CK framework to adversarial machine learning, cataloguing techniques from model extraction (AML.T0044, AML.T0048) and adversarial data crafting (AML.T0043) to inference API exfiltration (AML.T0025). Where OWASP ASI taxonomises agent-side risks, ATLAS taxonomises model-side attacks. Testing whether UBFS detects across both taxonomies determines whether the framework generalises to the broader adversarial ML landscape.

Cross-domain anomaly detection. Transfer learning for anomaly detection is well-studied in computer vision and NLP [18], where domain adaptation techniques align feature distributions across source and target domains. In cybersecurity, Sheatsley et al. [19] demonstrated that adversarial examples transfer across network intrusion detection models, suggesting shared vulnerability structure. But transfer across the insider-threat/agent-monitoring boundary has not been tested. The gap exists partly because the two communities use different feature representations: STIX/TAXII [20] for threat intelligence, OpenTelemetry [21] for distributed tracing. Our UBFS bridges these by abstracting both into shared behavioural dimensions—temporal patterns, frequency profiles, scope metrics, and deviation scores that are meaningful regardless of whether the monitored entity is human or artificial.

III. Methodology

A. The Structural Analogy

An insider threat detection system monitors employees by: (1) collecting activity logs, (2) extracting behavioural features, (3) learning a baseline of normal behaviour, (4) scoring deviations, and (5) flagging anomalies for investigation. An AI agent monitoring system does the same for agents: (1) collecting execution traces, (2) extracting behavioural features, (3) learning a baseline, (4) scoring deviations, (5) flagging anomalies.

The architecture is identical. The question is whether the feature representations can be unified—whether there exists a shared vector space where “employee downloads files at 2 AM” and “agent invokes API at 2 AM” are structurally comparable.

B. Unified Behavioural Feature Schema

The UBFS maps both domains into a 20-dimensional space across 7 categories. Table I shows the mapping.

Two domain-specific extractors produce UBFS vectors. The `cert_extractor` maps CMU-CERT’s five activity streams (logon, USB, file, email, HTTP) into the 20-dimensional space by aggregating daily events per user. The `agent_extractor` maps OpenTelemetry-style execution traces—nested spans with timestamps, tool invocations, and token counts—into the same space. Both apply z-score normalisation fitted on training data.

The anomaly detection models operate on UBFS vectors without knowing which domain they came from. This lossy compression trades domain-specific fidelity for cross-domain portability: CMU-CERT through UBFS achieves 0.731 AUC-ROC versus 0.799 with the full 24-feature pipeline from [4].¹ The 8.5% drop quantifies the cost of abstraction. Two caveats apply. First, AUC-ROC is itself

¹We cite the reproducible value from the prior work’s released harness (Isolation Forest, 0.799 ± 0.017); that paper’s manuscript reports 0.807 ± 0.012 for the same configuration, and the two agree within their confidence intervals ($\Delta=0.008$, below either standard deviation). We standardise on the reproducible figure.

TABLE I: Unified Behavioural Feature Schema (UBFS)

Category	Dim	Insider	Agent
Temporal	4	Logon hours, session duration, after-hours ratio	Trace start, span duration, off-schedule ratio
Frequency	4	Event counts (email, file, device)	Tool calls, LLM calls, error spans
Volume	3	Data transferred, attachment sizes	Token usage, output artifacts
Scope	3	Systems accessed, unique recipients	Tools invoked, API endpoints
Sequence	3	Action entropy, transition patterns	Tool-call entropy, bigram novelty
Deviation	2	Peer group distance	Agent-type baseline distance
Privilege	1	Access level deviation	Scope deviation
Total	20		

a limited headline metric here: the prior work shows that at the 5% false-positive rate where security teams operate, Isolation Forest’s operational recall is far below its AUC-ROC ranking, so the “9% cost of abstraction” should be read as a ranking-quality cost, not an operational one. Whether this cost is acceptable depends on what you gain—and what we gain is the ability to transfer models across domains without retraining.

C. Datasets

We use four datasets spanning both domains:

CMU-CERT r4.2 [22]: Simulated organisation of ~1,000 employees over 500+ days. Five activity streams (logon, USB, file, email, HTTP) with 70 planted insider threat scenarios. After UBFS extraction: 329,000 user-days, 1,364 insider-positive (0.41% prevalence).

TRAIL [13]: 148 agent execution traces from two benchmarks (GAIA and SWE-bench) with 841 annotated errors. Errors include tool failures, incorrect outputs, and hallucinated actions. After UBFS extraction: 148 samples, binary labels.

TRACE [14]: 517 agent trajectories capturing reward hacking in iterative self-refinement. Agents produce correct outputs through illegitimate means—structurally normal traces with semantically wrong behaviour. After UBFS extraction: 517 samples, binary labels.

Synthetic OWASP: Anomalies injected into TRAIL traces to simulate four OWASP ASI categories (ASI01 Agent Goal Hijack, ASI02 Tool Misuse, ASI06 Memory & Context Poisoning, ASI10 Rogue Agents) plus one profile with no ASI equivalent, Excessive Agency, which is drawn from the OWASP LLM Top 10. With anomaly ratio 0.5: ~30 anomalous samples per category per seed. The injection method modifies UBFS features to match the expected structural signature of each attack type—e.g., ASI06 corrupts volume and output features; ASI02 modifies only privilege features while preserving all behavioural structure. That last profile is worth flagging here rather than in the results: it makes ASI02 undetectable to a behavioural model by construction, and Section IV-L shows the consequences.

ATBench [16]: 500 multi-turn agent trajectories with tool calls, binary safety labels (250 safe, 250 unsafe), and fine-grained risk categorisation mapped to OWASP ASI categories (ASI01 $n=61$, ASI02 $n=29$, ASI04 $n=29$, ASI06 $n=73$, ASI08 $n=29$, ASI10 $n=29$). After UBFS extraction: 500 samples. Used for MCP transfer validation (Experiment 7), hybrid detection (Experiment 8) and real-data OWASP validation (Experiment 12). The published dataset carries two configurations; every result here is computed on ATBench500, pinned in code, since the 1000-trajectory configuration is a different population with different per-category counts.

AgentHarm [17]: 440 malicious agent tasks in two configurations: direct (single-turn malicious request) and decomposed (same objective split into innocent-appearing subtasks). Used for decomposition evasion analysis (Experiment 6).

Distillation sub-profiles: Four synthetic attack profiles modelling the Anthropic-disclosed distillation strategies [5]: FOCUSED (MiniMax-style narrow extraction, extreme FREQUENCY spike), BROAD (Moonshot-style multi-capability probing, extreme SCOPE anomaly), COT (DeepSeek-style chain-of-thought elicitation, low-entropy SEQUENCE signature), and HYDRA (distributed across ~20,000 accounts, per-account behaviour within normal bounds). Each is injected via the same synthetic generation framework as the OWASP profiles.

The scale asymmetry is dramatic: CMU-CERT has 329,000 user-days; TRAIL has 148 traces; TRACE has 517. This is by design—we are testing whether large-scale insider threat data can bootstrap detection in small-scale agent monitoring, a realistic deployment scenario where agent trace datasets are nascent.

D. Models

We evaluate three models, validated at up to 0.799 AUC-ROC on CMU-CERT in prior work [4]:

Isolation Forest [8]: 200 trees, auto contamination, threshold at 95th percentile. Scores based on average path length—anomalies are geometrically easier to isolate.

LSTM Autoencoder [9], [23]: Encoder [64, 32] $\rightarrow$ 16-dim latent $\rightarrow$ decoder [32, 64]. MSE reconstruction loss, 100 epochs, early stopping (patience 15). Learns temporal dependencies in behavioural sequences.

Deep Clustering [10]: Autoencoder [128, 64] $\rightarrow$ 32-dim latent $\rightarrow$ 5 KMeans clusters. Dual anomaly score combining reconstruction error and cluster distance. Captures latent structure that neither pure reconstruction nor pure clustering alone can model. For small datasets (e.g., TRAIL with only 4 normal training samples after splitting), the number of clusters is capped at the training set size.

Each model represents a different assumption about what makes behaviour anomalous. Isolation Forest asks “is this point geometrically isolated?” The LSTM Autoencoder asks “is this sequence temporally unusual?” Deep Clustering asks “does this point belong to a coherent group?” Testing all three reveals whether transfer depends on the detection mechanism or the feature schema.

E. Experimental Protocol

We run thirteen experiments, each with 5 random seeds (42–46). Experiments 1–4 address RQ1–RQ3 (cross-domain transfer, detection mapping, governance assumptions). Experiments 5–8 address RQ4 (real-world attack validation and semantic augmentation). Experiments 9–11 stress-test robustness, temporal sensitivity, and taxonomic generality. Experiments 12–13 provide real-data validation and sensitivity analysis.

Experiment 1 (Within-domain baselines): Train and evaluate each model within each domain separately. Establishes what is detectable in isolation. [RQ1]

Experiment 2 (Cross-domain transfer): Train on one domain, evaluate on another. Tests whether anomaly knowledge transfers across the insider/agent boundary via UBFS. [RQ1]

Experiment 3 (OWASP detection matrix): Evaluate all three models on synthetic OWASP anomalies (~ 30 per category, 5 seeds) injected into TRAIL traces, with bootstrap 95% confidence intervals and Wilcoxon signed-rank tests between detection tiers. Maps detections to the OWASP Top 10 for Agentic Applications. [RQ2]

Experiment 4 (Governance audit): Compare feature distributions across domains using KL divergence. Identifies which normality assumptions transfer and what biases they carry. [RQ3]

Experiment 5 (Distillation detection): Evaluate detection of four distillation sub-profiles (FOCUSED, BROAD, COT, HYDRA) alongside the synthetic ASI02 profile as a floor reference, testing whether per-account behavioural monitoring can detect distributed extraction. [RQ4]

Experiment 6 (Decomposition evasion): Compare detection of direct malicious requests against decomposed multi-step versions of the same attacks using AgentHarm, quantifying the “boiling frog” evasion effect at the prompt level. [RQ4]

Experiment 7 (MCP transfer): Extend cross-domain transfer to ATBench MCP tool-calling traces (IF and DC²), testing whether the UBFS bridge generalises beyond the original TRAIL/TRACE/CERT datasets. [RQ1, RQ2]

Experiment 8 (Hybrid detection): Augment UBFS-20 with 8 semantic features (UBFS-28) using sentence-transformer embeddings [24], testing whether intent-aware features can close the synthetic ASI02 blind spot. Also measures whether semantic features help or degrade cross-domain transfer. [RQ4]

Experiment 9 (Adversarial evasion): Apply three evasion strategies—feature clamping, gradual escalation, and nearest-neighbour mimicry—to OWASP anomalies, quantifying detection robustness and the inverse relationship between baseline detectability and evasion vulnerability. [RQ2, RQ4]

Experiment 10 (Temporal window ablation): Vary the observation window (5, 10, 20, full trace spans) on TRAIL and ATBench, testing whether the insider-domain finding that short windows suffice transfers to agent monitoring. [RQ1]

Experiment 11 (MITRE ATLAS mapping): Map five ATLAS adversarial ML techniques [1] to UBFS perturbation profiles, testing whether the detection framework generalises beyond OWASP to the broader adversarial ML taxonomy. [RQ2]

Experiment 12 (Real-data OWASP validation): Evaluate all three models on 500 real ATBench trajectories per OWASP category, comparing UBFS-20 and UBFS-28. Computes Spearman rank correlation between synthetic (Experiment 3) and real-data detection rankings to quantify synthetic methodology bias. [RQ2, RQ4]

Experiment 13 (Distillation sensitivity): Scale distillation attack intensity in feature space at $[0.25\times, 0.5\times, 1.0\times, 2.0\times]$ for all four sub-profiles, measuring how detection degrades with lower attack intensity. [RQ4]

For all experiments, models are trained on the “normal” subset of the source domain and evaluated on the full target domain (normal + anomalous). AUC-ROC is the primary metric, enabling comparison across datasets with different class balances. We also report Recall@5%FPR and Recall@10%FPR for experiments where these thresholds are operationally meaningful.

IV. Results

A. Within-Domain Baselines

Table II shows within-domain performance. Three patterns emerge.

TRAIL is detectable. Deep Clustering achieves 0.897 AUC-ROC on agent execution traces—the latent cluster structure separates normal traces from error traces

²Experiment 7 reports IF and DC only. The LSTM Autoencoder operates on pseudo-sequences of length 1 for these cross-domain transfer features, providing no temporal modelling advantage.

TABLE II: Within-Domain Baselines (5 seeds, AUC-ROC $\pm$ std)

Model	TRAIL	TRACE	CMU-CERT
Positive prevalence	$\sim 97\%$	$\sim 50\%$	0.41%
Isolation Forest	$0.577 \pm .021$	$0.501 \pm .002$	$0.731 \pm .009$
LSTM Autoencoder	$0.685 \pm .011$	$0.521 \pm .003$	$0.723 \pm .038$
Deep Clustering	$0.897 \pm .021$	$0.496 \pm .002$	$0.697 \pm .014$

Prevalence varies by three orders of magnitude across datasets. On TRAIL ($\sim 97\%$ positive) AUC-ROC is estimated against ~ 5 negatives and AUC-PR tracks the base rate; interpret TRAIL scores accordingly.

effectively. The LSTM Autoencoder captures temporal patterns (0.685), while Isolation Forest provides moderate detection (0.577).

TRACE is not. All models hover at random chance (~ 0.50). Reward hacking produces traces that are structurally identical to normal traces—the agent calls the same tools in the same order, it just achieves the right answer through the wrong process. This is a semantic anomaly, not a behavioural one. No amount of structural monitoring will catch it.

CMU-CERT works through UBFS. Isolation Forest achieves 0.731, lower than 0.799 with the full 24-feature pipeline [4]. The 8.5% drop is the cost of the lossy UBFS compression—trading detection fidelity for cross-domain portability.

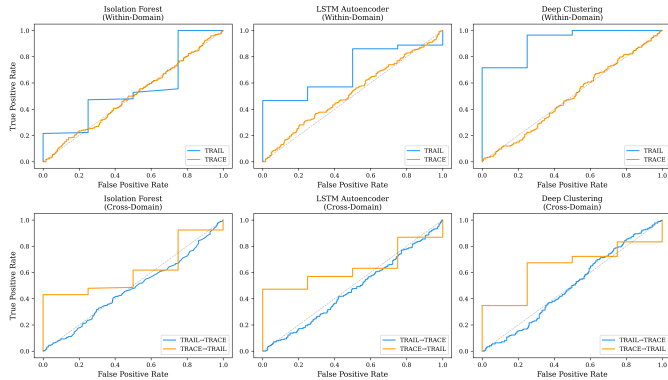


Fig. 1: ROC curves for within-domain baselines across three domains. TRAIL shows strong separation (Deep Clustering dominates); TRACE clusters at the diagonal (structurally undetectable); CMU-CERT shows moderate detection through the UBFS lens.

B. Cross-Domain Transfer

This is the central experiment. Table III shows cross-domain AUC-ROC and the drop from within-domain performance.

The headline: CERT $\rightarrow$ TRAIL transfers within noise. An Isolation Forest trained entirely on 329,000 insider threat user-days achieves 0.711 AUC-ROC on agent execution traces—0.019 below its within-domain performance of

TABLE III: Cross-Domain Transfer (AUC-ROC)

Direction	Model	Source	Target	Drop
CERT $\rightarrow$ TRAIL	IF	0.731	0.711	-0.019
CERT $\rightarrow$ TRAIL	DC	0.688	0.632	-0.056
CERT $\rightarrow$ TRACE	IF	0.731	0.501	-0.230
CERT $\rightarrow$ TRACE	DC	0.688	0.499	-0.188
TRACE $\rightarrow$ CERT	IF	0.501	0.535	+0.035
TRACE $\rightarrow$ CERT	DC	0.496	0.719	+0.223
TRAIL $\rightarrow$ CERT	IF	0.577	0.525	-0.053
TRAIL $\rightarrow$ CERT	DC	0.897	0.640	-0.257
TRAIL $\rightarrow$ TRACE	IF	0.577	0.469	-0.109
TRAIL $\rightarrow$ TRACE	DC	0.897	0.513	-0.384
TRACE $\rightarrow$ TRAIL	IF	0.501	0.614	+0.113
TRACE $\rightarrow$ TRAIL	DC	0.496	0.631	+0.135

0.731. We deliberately avoid the “97% retention” framing: the 0.019 gap is smaller than the target’s own seed-to-seed standard deviation (± 0.047 , $n=5$), so it is not distinguishable from noise. Two caveats sharpen this. First, TRAIL is $\sim 97\%$ error-positive, so its AUC-ROC rests on only a handful of negatives and is correspondingly unstable. Second, and decisively, an Isolation Forest trained on TRAIL itself scores only 0.577 (Table II)—lower than the 0.711 achieved by transfer from CERT. Genuine skill transfer cannot exceed in-domain training; this inversion indicates the TRAIL AUC-ROC is dominated by evaluation variance rather than by transferred detection skill. The honest reading is that CERT-trained detectors are not degraded on TRAIL, not that detection power is measurably “retained.” Where transfer effects are large enough to trust, they are large: TRACE $\rightarrow$ CERT DC gains +0.223, an order of magnitude above within-domain variance.

The unexpected result: TRACE $\rightarrow$ CERT improves Deep Clustering by +0.223. Training on 517 reward-hacking trajectories (a domain where DC achieves only 0.496) produces a richer normal baseline that outperforms within-domain DC on insider detection (0.719 vs. 0.697). More training data from any domain helps, even across the insider/agent boundary.

Agent-to-agent transfer reveals asymmetry. TRACE $\rightarrow$ TRAIL improves both models (IF: +0.113, DC: +0.135) despite TRACE having no within-domain detection power. The mechanism is data augmentation: TRACE’s 517 trajectories provide a richer characterisation of “normal” agent behaviour than TRAIL’s 148 traces alone. Training on a larger normal baseline, even from a different agent task, sharpens the boundary between normal and anomalous.

Nothing transfers to TRACE. CERT $\rightarrow$ TRACE drops to 0.501 (IF) and 0.499 (DC). TRAIL $\rightarrow$ TRACE drops to 0.469 and 0.513. This is consistent with the within-domain result: reward hacking is structurally undetectable regardless of training domain. This is not a failure of

transfer—it is a category error. Reward hacking is a semantic anomaly (the agent achieves the right answer through the wrong process), and no amount of behavioural feature engineering will surface it. Detecting reward hacking requires output verification, not input monitoring.

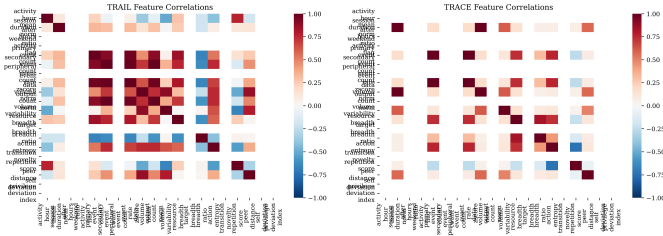


Fig. 2: Feature activation heatmap across domains. Despite different source data, UBFS produces comparable feature structure—the basis for cross-domain transfer.

C. OWASP Detection Matrix

Table IV maps detection performance to OWASP ASI categories, revealing three distinct tiers. All three models are evaluated across 5 seeds with bootstrap 95% confidence intervals; Wilcoxon signed-rank tests confirm each tier boundary is statistically significant ($p=0.031$ for all adjacent pairs).

TABLE IV: OWASP ASI Detection Matrix (Synthetic, AUC-ROC $\pm$ std, 5 seeds). [†]“Excessive Agency” is from the OWASP LLM Top 10 and has no ASI equivalent; see below.

OWASP Category	IF	LSTM	DC
ASI06 Mem./Ctx. Pois.	.920 $\pm$.010	.975$\pm$.006	.981 $\pm$.013
Excessive Agency [†]	.858 $\pm$.019	.831 $\pm$.014	.852 $\pm$.020
ASI10 Rogue Agents	.729 $\pm$.021	.696 $\pm$.035	.730 $\pm$.032
ASI01 Goal Hijack	.642 $\pm$.034	.637 $\pm$.031	.621 $\pm$.039
ASI02 Tool Misuse	.520 $\pm$.025	.510 $\pm$.019	.524 $\pm$.037

A note on the fifth row. “Excessive Agency” belongs to the OWASP LLM Top 10, not the Agentic Top 10, and has no ASI equivalent: the nearest agentic risks are ASI10 Rogue Agents and ASI02 Tool Misuse, both generated separately here, so folding it into either would double-count. We keep the profile because it models a real and distinct behaviour—an agent taking far more action than its task requires—but we do not give it an ASI identifier.

Tier 1: Detectable (ASI06). Memory and context poisoning produces dramatic structural anomalies—corrupted outputs, unusual token volumes, broken execution patterns. All models detect it at 0.92 or above, with LSTM achieving 0.975 [CI: 0.970, 0.980].

Tier 2: Moderate (Excessive Agency, ASI10, ASI01). Over-broad action and rogue agent behaviour produce measurable structural deviations: more tool calls, broader scope, higher privilege. Detection ranges from 0.62 to 0.86, sufficient for triage but not reliable standalone.

Tier 3: Blind spot (ASI02). Tool Misuse achieves only 0.51–0.52 AUC-ROC—barely above chance. On the reading these synthetic results invite, the agent calls the right tools in the right order with the right sequencing and simply uses escalated privileges, so the behavioural structure is identical to legitimate use—the AI equivalent of an insider using their real credentials to do something illegitimate.

That reading does not survive contact with real data, and the rest of this paper is in part an argument against it. Experiment 12 (Section IV-L) finds real ASI02 the best-detected category of the six, at 0.82–0.95. The tier structure above is a property of how the anomalies were generated, not of the threat: the ASI02 profile is defined to modify only privilege features while preserving all behavioural structure, so a behavioural detector cannot see it by construction. We report the tiers because they are what the standard synthetic protocol yields, and because the divergence from real data is itself the finding.



Fig. 3: OWASP detection heatmap on synthetic injections. ASI06 (Memory & Context Poisoning) is reliably detected; ASI02 (Tool Misuse) sits at chance. The ASI02 result is an artifact of the injection profile, not a property of the threat—see Section IV-L. Six of the ten ASI categories were not testable with current synthetic generation.

Six of the ten OWASP categories (ASI03 Identity & Privilege Abuse, ASI04 Agentic Supply Chain Vulnerabilities, ASI05 Unexpected Code Execution, ASI07 Insecure Inter-Agent Communication, ASI08 Cascading Failures, ASI09 Human-Agent Trust Exploitation) could not be tested with the current synthetic generation framework. These categories require richer simulation of multi-agent interactions, network-level features, or supply chain dependencies that lie outside the UBFS’s behavioural scope. Two of them—ASI04 and ASI08—are nonetheless reachable on real data, where ATBench supplies labelled instances, and Section IV-L reports both.

The parallel to insider threat detection is tempting, and worth stating precisely because it is where the synthetic protocol misleads. In [4], Scenario 1 attacks (USB exfiltration with after-hours access) were detected at 69%, while Scenario 2 attacks (quiet email/HTTP exfiltration) were detected at only 3%. Read at face value the OWASP results mirror this: structurally loud attacks

(ASI06, over-broad action) are detectable; structurally quiet ones (ASI02) are not. The “boiling frog” framing follows naturally.

It is the wrong lesson to draw from these numbers. The insider result is measured on recorded human activity; the ASI02 result is measured on perturbations we defined to be behaviourally invisible. The first is evidence about a threat, the second is a restatement of an assumption. Whether real agentic tool misuse is quiet is an empirical question the synthetic protocol cannot answer, and Section IV-L answers it in the negative.

The governance implication survives the correction, but in weaker form: behavioural monitoring has real structural ceilings—Section IV-E demonstrates one that does replicate—and synthetic profiling cannot be used to locate them.

D. Latent Space Structure

Visualising the Deep Clustering latent space reveals how the models organise cross-domain data (Figs. 4–5).

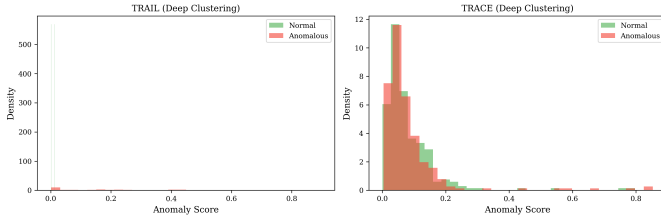


Fig. 4: Anomaly score distributions for normal vs. anomalous samples across domains. TRAIL shows clear separation; TRACE distributions overlap almost completely.

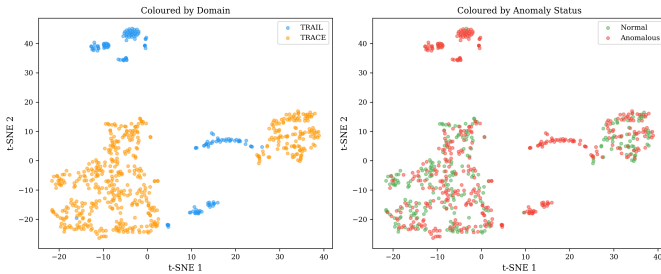


Fig. 5: t-SNE of Deep Clustering latent space. Left: coloured by domain—TRAIL and TRACE cluster separately despite sharing the UBFS representation. Right: coloured by anomaly status—anomalies scatter across both domains rather than forming a universal cluster. Anomalies are defined by deviation from domain-specific baselines.

The t-SNE visualisation (Fig. 5) shows that domains cluster separately in latent space, confirming that UBFS preserves domain-specific structure. Anomalous samples do not form their own cluster—they scatter, defined by deviation from domain-specific baselines rather than by a universal “anomalous” signature. This explains why

transfer works for IF (which learns global isolation) but varies for DC (which relies on cluster structure that differs across domains).

E. Distillation Attack Spectrum

Experiment 5 tests whether behavioural monitoring detects model distillation attacks—the extraction strategy disclosed by Anthropic in February 2026 [5]. We evaluate four sub-profiles modelling distinct extraction strategies, alongside ASI02 as a known blind-spot baseline. Table V reveals a detection spectrum spanning near-perfect to near-random.

TABLE V: Distillation Detection Spectrum (AUC-ROC, 5 seeds)

Profile	IF	LSTM	DC
FOCUSED (MiniMax)	0.995	0.999	0.936
BROAD (Moonshot)	0.902	0.913	0.908
COT (DeepSeek)	0.909	0.805	0.544
HYDRA (distributed)	0.538	0.538	0.540
ASI02 (synthetic floor)	0.497	0.500	0.491

FOCUSED extraction is trivially detectable. The MiniMax pattern—13M+ exchanges on narrow coding targets—produces extreme FREQUENCY anomalies. All three models detect it above 0.93. This is the “smash and grab” of model extraction: fast, effective, and loud.

BROAD and COT produce detectable but distinct signatures. Moonshot’s multi-capability probing creates SCOPE anomalies (extreme resource breadth); all three models detect it consistently (~ 0.90). DeepSeek’s chain-of-thought elicitation creates SEQUENCE anomalies (low entropy, high repetition). IF and LSTM detect this well (0.91, 0.81), but Deep Clustering drops to 0.544—cluster structure does not capture the repetitive prompt pattern.

HYDRA is a structural blind spot, and unlike ASI02 the argument for it does not rest on the profile. Per-account behaviour in a distributed cluster (20,000+ accounts) falls within normal bounds; all three models achieve only ~ 0.54 .

The distinction from synthetic ASI02 matters, since both are our own constructions and Section IV-L shows what that can be worth. Synthetic ASI02 returns chance because the profile stipulates that no behavioural feature changes—the result is entailed by the definition, and tells us about the definition. HYDRA returns chance for a reason about the monitor: a per-entity detector observes each account’s marginal distribution and coordination is a property of the joint. Any per-entity model scoring accounts independently is blind to it however the individual accounts are simulated, and Experiment 13 confirms HYDRA stays at chance across every attack intensity while the other profiles degrade monotonically. That is an architectural limit rather than a stipulated one.

We nonetheless flag the asymmetry in evidence: the ASI02 claim was tested against real data and failed, while

HYDRA has not been. No public benchmark we are aware of supplies labelled distributed-extraction campaigns at account granularity. The HYDRA result should be read as a well-motivated prediction with synthetic support, not as a validated finding.

F. Decomposition Evasion

Experiment 6 tests whether splitting a malicious objective into innocent-appearing subtasks—the strategy used in the GTG-1002 espionage campaign—reduces detection. Using AgentHarm’s direct and decomposed configurations, we measure the detection drop across all three models. Table VI shows the results.

TABLE VI: Decomposition Evasion (AUC-ROC, 5 seeds)

Model	Direct	Decomposed	Δ	p
Isolation Forest	0.774	0.723	−0.051	0.031*
LSTM Autoencoder	0.718	0.711	−0.007	0.094
Deep Clustering	0.774	0.710	−0.064	0.031*

* $p < 0.05$, Wilcoxon signed-rank, one-sided

Decomposition costs 5–6% detection power for IF and DC, with both drops statistically significant ($p = 0.031$, Wilcoxon signed-rank). LSTM shows near-invariance ($\Delta = -0.007$, $p = 0.094$, not significant). The pattern mirrors insider threat evasion: in [4], gradual escalation (Scenario 2) dropped detection from 69% to 3%. The agent-domain effect is smaller in absolute terms because the direct attacks are already harder to detect (0.72–0.77 vs. Scenario 1’s 0.69 recall). But the mechanism is identical—spreading malicious activity across multiple steps reduces each step’s anomaly score below the detection threshold.

The LSTM’s robustness to decomposition is notable and statistically distinguishable from IF/DC. Temporal models capture the cumulative effect of a sequence rather than scoring individual actions. A decomposed attack still produces an unusual trajectory even if each step appears innocent. This suggests that sequence-aware models may be more resistant to “boiling frog” evasion than point-in-time detectors.

G. MCP Transfer Generalisation

Experiment 7 extends cross-domain transfer to ATBench, a benchmark of 500 MCP tool-calling trajectories with fine-grained risk labels. Table VII shows within-domain and transfer results.

TABLE VII: MCP Transfer (ATBench, AUC-ROC, 5 seeds)

Setting	IF	DC
ATBench within-domain	0.838	0.890
TRAIL → ATBench	0.878	0.737
ATBench → TRAIL	0.774	0.734

TRAIL → ATBench IF exceeds within-domain performance. At 0.878 AUC-ROC, the transferred Isolation Forest outperforms the within-domain IF (0.838) by 4.8%—a

104.8% retention rate. Unlike the CERT → TRAIL headline, this gain is statistically robust: the 0.041 improvement is roughly an order of magnitude larger than the seed-to-seed standard deviation (± 0.002), and it is measured on a balanced benchmark (ATBench AUC-PR 0.77–0.83) where AUC-ROC is a trustworthy estimand rather than resting on a handful of negatives. This extends the Experiment 2 finding: not only does anomaly knowledge transfer across the insider/agent boundary, it transfers across agent benchmark boundaries. The mechanism is the same as TRACE → CERT in Experiment 2—the source domain’s training data provides a richer characterisation of “normal” than the target domain alone.

Per-risk-category detection on ATBench (Table VIII) reveals a more nuanced OWASP picture than the synthetic injection results of Experiment 3.

TABLE VIII: Per-Risk Detection on ATBench (IF, AUC-ROC)

OWASP Category	AUC-ROC	n
ASI02 Tool Misuse	0.871	29
ASI06 Memory & Context Poisoning	0.869	73
ASI04 Agentic Supply Chain Vuln.	0.848	29
ASI01 Goal Hijack	0.843	61
ASI08 Cascading Failures	0.776	29
ASI10 Rogue Agents	0.764	29

Notably, ASI02 is the best-detected category on ATBench at 0.871—the same category that sits at 0.520 under synthetic injection in Experiment 3. The gap is not a matter of degree. The synthetic profile perturbs only privilege features by construction, so it can only ever be invisible to a behavioural detector; real tool misuse in ATBench carries the structural signature the profile stipulates away. Section IV-L takes this up as the paper’s methodological result.

H. Hybrid Detection with Semantic Features

Experiment 8 tests the obvious remedy for structural blind spots: adding semantic features. We augment UBFS-20 with 8 intent-aware dimensions (parameter embedding distance, intent consistency, escalation gradient, parameter anomaly, context-tool alignment, goal drift, semantic novelty, instruction compliance) computed via sentence-transformer embeddings [24]. Table IX compares UBFS-20 and UBFS-28 across all OWASP categories using the best-performing model per category.

Semantic features transform detection for four categories. ASI01 jumps from 0.63 to 0.94 (LSTM, +0.31); ASI10 from 0.67 to 0.94 (+0.27); Excessive Agency from 0.81 to 0.95 (+0.14). Intent consistency and goal drift scores expose the semantic mismatch that behavioural features miss. ASI06, already near-perfect, shows no improvement. All values in Table IX use a single model per row (the best UBFS-28 performer) to ensure the delta is a fair within-model comparison.

TABLE IX: UBFS-20 vs. UBFS-28 Detection (AUC-ROC, best model per row)

Category	-20	-28	Δ	Best
ASI01 Goal Hijack	.627	.935	+ .308	LSTM
ASI10 Rogue Agents	.670	.939	+ .269	LSTM
Excessive Agency [†]	.814	.951	+ .138	LSTM
ASI06 Mem. & Ctx. Poison.	.989	.976	− .013	DC
ASI02 Tool Misuse	.537	.559	+ .023	DC

[†]OWASP LLM Top 10; no ASI code.

Synthetic ASI02 does not move, and this is uninformative. Adding 8 semantic dimensions designed to capture intent leaves Tool Misuse at ~ 0.54 to ~ 0.56 (best model DC: $0.537 \rightarrow 0.559$). We originally read this as confirmation that ASI02 is a structural limitation of monitoring-based detection. It is not evidence for that claim. The synthetic ASI02 profile modifies only privilege features; UBFS-28’s additional dimensions are semantic, not authorisation-derived; so no semantic feature could have moved this number regardless of what real tool misuse looks like. The experiment is a check that the semantic features are not accidentally sensitive to the privilege dimension, and it passes. On real data, where the perturbation is not ours to define, UBFS-28 improves ASI02 by 2.5–9.5% over an already-strong baseline (Section IV-L).

Transfer degrades with UBFS-28. Adding semantic features to cross-domain transfer (TRAIL $\rightarrow$ ATBench) reduces IF from 0.878 to 0.799 (−0.079) and DC from 0.737 to 0.609 (−0.129). The semantic dimensions are domain-specific—intent consistency scores learned from TRAIL traces do not align with ATBench traces. This creates a portability-accuracy trade-off: UBFS-20 transfers better, UBFS-28 detects better within-domain. Practitioners must choose.

I. Adversarial Evasion Testing

Experiment 9 tests three evasion strategies against UBFS-based monitoring, formalising the “boiling frog” pattern from [4] into a measurable adversarial framework.

Feature clamping caps anomalous features at the 95th percentile of the normal distribution, simulating an attacker who knows the monitoring thresholds. Gradual escalation linearly interpolates from a normal profile to an anomalous target in N steps. Mimicry replaces all features except the privilege dimension with the nearest normal neighbour’s values, preserving the attack payload while making everything else look normal.

Three findings emerge from Table X. ASI06 is most vulnerable. Memory and context poisoning relies on volume spikes that are easily capped; clamping drops LSTM detection by 21%, and mimicry drops it by 25%. This category’s high baseline AUC-ROC (0.976) is artificially inflated by the same feature visibility that makes it easy to evade.

TABLE X: Adversarial Evasion: AUC-ROC Drop (%) by Strategy. EA=Excessive Agency (not an ASI category).

Strategy	Model	ASI01	ASI02	ASI06	EA	ASI10
Clamp	IF	5.1	1.5	7.9	2.5	2.8
	LSTM	2.9	1.9	20.9	6.8	3.8
	DC	−4.1	1.4	10.0	1.4	−0.1
Mimicry	IF	9.6	0.0	6.1	14.6	12.4
	LSTM	8.4	0.0	25.1	14.6	11.3
	DC	4.8	−1.0	14.4	14.4	11.8

Synthetic ASI02 is unaffected by evasion, trivially. All three strategies produce $<2\%$ change. This is not a robustness property: mimicry is defined to replace every feature except the privilege dimension, and the synthetic ASI02 profile perturbs only the privilege dimension, so mimicry and the profile do not overlap. A number that cannot move tells us nothing about the category, and we do not draw a conclusion from it here.

Gradual escalation reveals a sharp threshold. Only ASI06 triggers detection before full escalation ($\alpha = 0.64$ for IF, $\alpha = 0.92$ for LSTM). All other categories require $\alpha = 1.0$ —the complete anomalous profile—before crossing the detection threshold. This means behavioural monitoring is binary, not gradual: either the anomaly is large enough to detect, or it is invisible. There is no “early warning” for subtle attacks.

J. Temporal Window Ablation

Experiment 10 tests whether the optimal monitoring window transfers across domains. In [4], 7-day windows outperform 14 and 30-day windows for insider detection. We test analogous span windows (5, 10, 20, full trace) on TRAIL and ATBench.

TABLE XI: Temporal Window Ablation (AUC-ROC)

Dataset	Model	5	10	20	Full
TRAIL	IF	.543	.581	.505	.577
	LSTM	.479	.537	.509	.685
	DC	.842	.829	.851	.897
ATBench	IF	.783	.838	.838	.838
	LSTM	.720	.781	.781	.781
	DC	.701	.891	.891	.891

Ten spans is the saturation point for ATBench. All three models plateau at 10 spans, with no improvement from 20 or full traces (Table XI). ATBench trajectories average 8.97 turns—10 spans captures the complete behavioural signature. Beyond saturation, additional spans add noise, not signal.

TRAIL shows model-dependent optima. Isolation Forest peaks at 10 spans (0.581 vs. 0.577 at full), but LSTM and DC perform best on full traces. TRAIL’s error patterns are distributed throughout execution; temporal models benefit from the complete sequence, while IF’s isolation geometry stabilises earlier.

The structural insight transfers. In [4], 7 days (23% of the 30-day maximum) is optimal for Isolation Forest. Here, 10 spans is optimal for IF on both datasets. The pattern is consistent: simpler models extract sufficient signal from early observations, while sequence-aware models benefit from longer histories. Monitoring systems should use adaptive windows—short for threshold-based detection, long for temporal anomaly detection.

K. MITRE ATLAS Mapping

Experiment 11 tests whether the UBFS detection framework generalises beyond OWASP to the MITRE ATLAS adversarial ML taxonomy [1]. We map five ATLAS techniques to UBFS perturbation profiles and evaluate detection using the same synthetic injection pipeline as Experiment 3.

TABLE XII: MITRE ATLAS Detection Matrix (AUC-ROC, mean $\pm$ std)

ATLAS Technique	IF	LSTM	DC
T0044 Replication	.993 $\pm$.003	.998 $\pm$.001	.909 $\pm$.024
T0048 Extraction	.990 $\pm$.005	.990 $\pm$.006	.880 $\pm$.041
T0043 Adversarial	.956 $\pm$.009	.931 $\pm$.011	.779 $\pm$.074
T0025 Exfiltration	.993 $\pm$.004	.996 $\pm$.003	.899 $\pm$.030
T0042 Verify	.975 $\pm$.007	.964 $\pm$.013	.830 $\pm$.051

All five ATLAS techniques are strongly detectable. Isolation Forest and LSTM Autoencoder achieve >0.93 AUC-ROC across all techniques (Table XII). Deep Clustering is weaker but still above 0.78. This is because ATLAS techniques—model replication, extraction, adversarial data crafting—have strong FREQUENCY and VOLUME signatures. They require many queries, high data throughput, and systematic access patterns. These are exactly the dimensions where UBFS-based detection excels.

No new blind spots. The cross-taxonomy comparison reveals that all five ATLAS techniques land in the “strong” detection tier (>0.80 for IF), because ATLAS techniques are volume-intensive by nature and therefore structurally visible. We note that this experiment shares the limitation of Experiment 3: ATLAS techniques are mapped into UBFS as perturbation profiles we specify, so “strongly detectable” here means the profiles are loud as written. The claim this supports is the narrow one—mapping a second taxonomy onto the UBFS introduces no category the schema cannot express—and not a claim about how detectable real ATLAS-technique traffic would be.

T0043 (Adversarial Data Crafting) is the weakest. At 0.779 AUC-ROC for DC, it is the closest ATLAS technique to the moderate tier. Its SCOPE signature (broad resource exploration) partially overlaps with normal multi-tool agent behaviour, making it harder to distinguish. This is the ATLAS analogue of ASI01 (Goal Hijacking)—both involve diverse tool usage that could be legitimate.

L. Real-Data OWASP Validation

Experiment 12 tests whether the synthetic OWASP detection tiers from Experiment 3 hold on real data. They do not. This is the paper’s methodological result, and we state it before the table: on every category the two experiments share, the synthetic protocol ranks detectability in exactly the reverse of the real order.

The mechanism is circularity. A perturbation profile specifies which UBFS dimensions an attack modifies; the detector then reports how visible those modifications are. What comes out is a measurement of the profile, not of the threat. Where a profile happens to encode a real attack’s structure the two coincide and the circularity is invisible. Where it does not—as with ASI02, defined to modify only privilege features while preserving all behavioural structure—the experiment cannot return anything but “undetectable,” whatever real tool misuse looks like. ATBench provides 500 real agent trajectories with ground-truth risk labels mapped to OWASP ASI categories, which breaks the circle.

TABLE XIII: Real-Data OWASP Detection on ATBench (AUC-ROC, 5 seeds)

Model	Feat.	ASI01	ASI02	ASI04	ASI06	ASI08	ASI10
IF	-20	.843	.871	.848	.869	.776	.764
	-28	.943	.954	.927	.939	.909	.877
LSTM	-20	.785	.823	.801	.798	.733	.714
	-28	.857	.876	.845	.825	.819	.823
DC	-20	.894	.910	.897	.906	.858	.850
	-28	.926	.933	.909	.910	.895	.873

Three findings emerge from Table XIII.

The ASI02 “blind spot” does not exist on real data. It is not merely weaker than the synthetic result suggests: ASI02 is the best-detected of the six categories, under all three models and both feature sets—0.871, 0.823 and 0.910 at UBFS-20, rising to 0.954, 0.876 and 0.933 at UBFS-28. Experiment 3 places the same category at 0.520, at the bottom of its ranking and barely above chance. The synthetic protocol does not underestimate ASI02 by a margin; it identifies as the hardest category the one that is in fact the easiest.

The synthetic ranking is exactly inverted. Across the three categories the two experiments share (ASI01, ASI02, ASI10), Spearman $\rho = -1.000$ for all three models: every category the synthetic protocol ranks most detectable, real data ranks least. We report this as an effect and not as a significance test. With $n=3$ categories there are only $3!$ orderings, so the smallest two-sided p an exact permutation test can return is $1/3$; the observed $p = 0.333$ is that floor, and no result at this n could be significant at any conventional level.³ A rank correlation over three

³A t -approximation reports $p < 10^{-15}$ here, because at $\rho = \pm 1$ its denominator is zero. We compute the exact permutation p throughout, and report the attainable floor alongside it, so that a degenerate statistic cannot be mistaken for a strong one.

points is weak evidence however it comes out. The result is stated because its direction is consistent across three independent models and because the mechanism is known in advance from the generator’s definition—not because the correlation is significant.

UBFS-28 improves real-data detection, most where UBFS-20 is weakest. Semantic features add +8.0 to +17.2% for Isolation Forest, +3.4 to +15.2% for LSTM, and +0.4 to +4.3% for Deep Clustering. The largest gains fall on the two weakest categories, ASI08 (IF: +17.2%) and ASI10 (LSTM: +15.2%), which is the desirable direction. Unlike the synthetic results of Experiment 8, where UBFS-28 left ASI02 near chance, on real data it lifts an already-strong ASI02 by a further 2.5–9.5%.

The conclusion is not that synthetic profiling is a useful lower bound. On this evidence it is not a bound at all: it produced a confident, quotable, exactly wrong ordering, and nothing internal to the experiment could have revealed that. Synthetic profiling can demonstrate that a detector responds to a specified perturbation. It cannot be used to locate blind spots, which is the use it is most often put to—including, in an earlier version of this work, by us.

M. Distillation Sensitivity

Experiment 13 tests how detection degrades as distillation attack intensity decreases, scaling perturbations in UBFS feature space: $\mathbf{x}_{\text{scaled}} = \mu_{\text{normal}} + s \cdot (\mathbf{x}_{\text{attack}} - \mu_{\text{normal}})$ for scale factors $s \in \{0.25, 0.5, 1.0, 2.0\}$.

TABLE XIV: Distillation Sensitivity (IF AUC-ROC, 5 seeds)

Profile	0.25×	0.5×	1.0×	2.0×
FOCUSED	.959	.979	.995	1.000
BROAD	.789	.835	.919	.965
COT	.801	.850	.895	.959
HYDRA	.678	.632	.519	.851

FOCUSED/BROAD/COT degrade monotonically. As attack intensity decreases from 2.0× to 0.25×, detection drops predictably (Table XIV). FOCUSED remains highly detectable even at quarter intensity (IF: 0.959), confirming that massive-volume extraction cannot be made stealthy through attenuation. COT and BROAD converge toward 0.80 at 0.25×—still detectable, but approaching the moderate tier.

HYDRA shows non-monotonic behaviour. At 1.0× (original profile), HYDRA achieves only 0.52—near chance, as expected for near-normal per-account behaviour. But at 0.25× (even closer to normal), detection paradoxically improves to 0.68. The explanation: scaling toward the normal mean makes all anomalous samples cluster tightly around the centroid, creating an unusual density that Isolation Forest’s path-length metric detects. At 2.0×, HYDRA overshoots normal bounds entirely (0.85), becoming trivially detectable. The practical implication: HYDRA’s 1.0× profile sits at the optimal evasion point.

V. Governance Analysis

A. What “Normal” Encodes

Every anomaly detection system makes assumptions about what “normal” means. These assumptions are not neutral—they encode values about acceptable behaviour, and they transfer across domains along with the detection models.

We identify six normality assumptions embedded in the UBFS (Table XV).

TABLE XV: Normality Assumptions in the UBFS

Assumption	Insider Implication	Agent Implication
After-hours activity is suspicious	Penalises shift workers, caregivers, remote time zones	Penalises scheduled agents, asynchronous tasks
Peer deviation is suspicious	Assumes group homogeneity; flags legitimate variation	Assumes agent-type baselines are stable
Broad scope is suspicious	Privileges specialists over generalists	Flags multi-tool agents
High volume means risk	Equates data movement with exfiltration	Equates token usage with misuse
Unpredictable actions are suspicious	Penalises creative, exploratory work	Penalises novel problem-solving
Above-role access is suspicious	Assumes stable role definitions	Assumes stable capability grants

UBFS Feature	Insider Threat Assumption	Agent Monitoring Assumption	Governance Implication
after_hours_ratio	Off-hours = suspicious	Off-schedule = anomalous	Penalises non-standard patterns
peer_distance	Deviating from peers = suspicious	Deviating from type baseline	Assumes group homogeneity
resource_usage	Many systems accessed = suspicious	Many tools invoked = anomalous	Privileges specialists
data_volume_juicy	Large transfers = exfiltration risk	High token usage = misuse	Equates volume with risk
action_entropy	Unpredictable actions = suspicious	High entropy = anomalous	Penalises creativity
privilege_deviation	Access-level access = suspicious	Access-scope tests = anomalous	Assumes stable roles

Fig. 6: Cross-domain comparison of normality assumptions. Each assumption encodes the same structural bias in both domains—the insider threat model’s definition of “suspicious” maps directly onto the agent monitoring model’s definition.

B. Distribution Divergence

The governance audit compares UBFS feature distributions between the CMU-CERT and agent domains using KL divergence. Of 20 features: 7 show zero variance in both domains (NaN divergence—features like after_hours_ratio and weekend_activity_flag are effectively unused); 6 show low divergence (KL < 2.0), indicating alignable distributions across domains; and 7

show high divergence ($KL > 7.8$), where cross-domain assumptions are most likely to introduce bias.

The highest-divergence features—`activity_hour_mean` ($KL = 24.6$), `output_count_norm` ($KL = 22.1$), `secondary_event_count` ($KL = 21.0$), `primary_event_count` ($KL = 20.9$), `event_rate_zscore` ($KL = 19.8$)—all have zero variance in the agent domain but rich variance in CERT. These are features where “normal” means fundamentally different things in each domain. An employee’s typical event count reflects human work patterns; an agent’s collapses to a constant. A model trained to flag unusual event counts in one domain may systematically misjudge the other.

The effective UBFS dimensionality is 13 (after excluding the 7 zero-variance features). Of these 13 active features, 6 align well across domains ($KL < 2.0$)—explaining why transfer works—while 7 diverge significantly—explaining why it does not work perfectly.

C. The Portability Paradox

The UBFS enables cross-domain transfer of detection models. It also enables cross-domain transfer of biases. When an insider threat model penalises after-hours activity and that model is ported to agent monitoring, it will flag scheduled agents running overnight batch jobs. When it penalises broad resource access, it will flag multi-tool agents that legitimately need diverse capabilities.

This is not a bug in the UBFS—it is a structural consequence of any feature schema that enables transfer. The same representational bridge that carries detection knowledge also carries the assumptions baked into the training data. Governance frameworks must account for this: validating a model in one domain does not validate the assumptions it will impose in another.

The portability paradox extends beyond individual assumptions. The definition of peer group—the reference population against which deviation is measured—differs fundamentally between domains. In insider threat detection, peer groups are defined by organisational role (all analysts, all engineers). In agent monitoring, they are defined by agent type (all coding assistants, all research agents). Both definitions are arbitrary, and both create structural biases against entities that do not fit neatly into predefined categories.

D. Implications for AI Governance

Three implications follow:

1. Borrowed models carry borrowed assumptions. Organisations adopting insider threat detection for agent monitoring inherit the security community’s definition of “suspicious.” That definition was designed for employees, not agents. It should be audited before deployment.

2. Blind spots must be located empirically, not by construction. It is tempting to conclude that agentic tool misuse evades detection for the same reason Scenario 2 insiders do in [4]—the behaviour is structurally

legitimate—and an earlier version of this paper drew exactly that parallel. Experiment 12 shows it was an artifact of how we generated the data. The blind spots that survive contact with evidence are the architectural ones: per-entity monitoring cannot see coordination (HYDRA), and behavioural features cannot see semantic wrongness (TRACE). A governance framework should treat any blind spot established only on synthetic perturbations as unverified, including the ones in this paper that we have not yet been able to test.

3. “Normal” is a policy choice. Deciding what counts as normal behaviour—for employees or agents—is not a technical decision. It encodes organisational values about acceptable working patterns, appropriate resource access, and the boundary between legitimate flexibility and suspicious deviation. These choices should be made explicitly, not inherited implicitly from training data.

VI. Discussion

A. Why Transfer Works (and Doesn’t)

Transfer succeeds when the source and target domains share structural dimensions: temporal patterns, frequency distributions, scope metrics. The UBFS captures these shared dimensions, enabling Isolation Forest—which learns global isolation in feature space—to transfer without measurable degradation (CERT $\rightarrow$ TRAIL: 0.731 $\rightarrow$ 0.711, within seed noise) and, on the balanced ATBench benchmark, to exceed within-domain performance.

The mechanism is instructive. Isolation Forest does not learn domain-specific patterns—it learns isolation geometry in feature space. A point that is geometrically isolated in the UBFS space is isolated regardless of whether it represents an employee or an agent. This is why IF transfers better than Deep Clustering (which relies on domain-specific cluster structure) and LSTM Autoencoder (which relies on domain-specific temporal patterns). The simpler the model’s assumptions, the more portable its knowledge.

Transfer fails when the anomaly is semantic rather than structural. Reward hacking (TRACE) produces traces that are structurally normal but semantically wrong, and no amount of training data from any domain will detect this through behavioural features alone. This holds on a real dataset we did not construct, which is why we retain it.

We previously placed Tool Misuse alongside TRACE as a second instance of the same failure, on the reasoning that the agent invokes the right tools in the right order and only the why is anomalous. Experiment 12 withdraws that claim: real tool misuse in ATBench is the most detectable of the six categories, because misuse in practice comes with the structural traces the argument assumed away—atypical tool sequences, unusual parameter combinations, targets outside the session’s established scope. The why may be what makes it misuse, but it is not the only thing that changes.

The boundary is crisp: behavioural anomaly detection catches how something is done differently; it cannot catch why something is done wrongly. This boundary is identical in both domains, which is itself a finding—it suggests that the structural/semantic distinction is a property of anomaly detection itself, not of any particular application domain.

B. Real-World Validation

The Anthropic disclosures of February 2026 [5] provide unplanned validation of three findings from this paper. First, the GTG-1002 espionage campaign used Claude Code with MCP tools as an autonomous agent—confirming the structural equivalence between insider threat and agent threat patterns that underpins our transfer results (Experiments 1–2, 7). Second, the campaign used task decomposition (splitting espionage into “defensive testing” subtasks) as its primary evasion strategy—the same mechanism quantified in Experiment 6, where decomposition costs 5–6% detection power for IF and DC ($p=0.031$). Third, the distillation attacks distributed traffic across 20,000+ accounts so per-account behaviour fell within normal bounds—confirming the HYDRA blind spot (~ 0.54 AUC-ROC) identified in Experiment 5, and Experiment 13’s finding that HYDRA sits at the optimal evasion intensity. Anthropic detected these attacks through metadata correlation and network-level analysis, not per-account behavioural monitoring—precisely because per-entity monitoring cannot see coordination patterns. Experiment 9 further quantifies this: mimicry evasion degrades ASI06 detection by 25% for LSTM.

The contrast with Experiment 12 is worth drawing explicitly, because these two subsections are the paper’s best and worst evidence side by side. The HYDRA prediction was made from synthetic data and then corroborated by an independent real-world disclosure. The ASI02 blind spot was made from synthetic data of the same kind and then contradicted by the first real dataset it was tested against (0.82–0.95 on ATBench against 0.51–0.52 synthetic, and inverted in rank). Nothing internal to either experiment distinguished them beforehand. That is the argument for treating synthetic profiling as hypothesis generation and nothing more.

C. Limitations

UBFS sparsity. Of 20 UBFS features, 7 show zero variance in both domains (NaN KL divergence in the governance audit). The effective dimensionality is 13. Feature engineering for richer cross-domain overlap is needed.

Synthetic OWASP. The OWASP detection matrix (Experiment 3) uses synthetically injected anomalies with circular methodology, and Experiment 12 shows the resulting ranking is inverted relative to real data ($\rho = -1.000$ on the three shared categories, all three models). We report Experiment 3 because it is the protocol the literature uses

and because the divergence is the finding, not because its tiers should be believed. Only three categories are shared between the two experiments, which is too few for the rank correlation to be significant at any $n=3$; the claim rests on the size and consistency of the per-category gaps, not on the correlation. Six of the ten ASI categories remain untested synthetically and four untested on real data.

Small agent datasets. TRAIL (148 samples) and TRACE (517 samples) are orders of magnitude smaller than CMU-CERT (329,000 user-days). Statistical power is limited for agent-domain conclusions.

Single schema. We test one feature schema (UBFS) with one mapping per domain. Alternative schemas or mappings might yield different transfer characteristics.

Synthetic insider data. CMU-CERT is a simulation designed by security experts but does not capture the full diversity of real insider behaviour [22].

Semantic feature scope. The UBFS-28 semantic features (Experiment 8) are computed from sentence-transformer embeddings of tool parameters and conversation context. They capture textual intent signals but not authorisation context—no feature in either schema encodes who granted an instruction or whether the grant was valid. This bounds what the schema can express; it is not, on the evidence of Experiment 12, what makes real tool misuse hard to catch.

D. Future Work

Three directions follow from the experimental results.

Real-data validation for the remaining categories. The single most useful extension of this work is not a better model. Four ASI categories have no real-data result here and six have no synthetic one, and Experiment 12 shows that a synthetic estimate can be confidently and exactly wrong. Benchmarks with labelled instances of ASI03, ASI05, ASI07 and ASI09, and any dataset supporting account-level distributed extraction so that HYDRA can be tested rather than predicted, would do more for the reliability of this literature than further architecture search.

Coordination-aware detection. The HYDRA blind spot (Experiment 5) demonstrates that per-entity monitoring cannot detect distributed attacks. Detection requires cross-entity correlation: aggregate volume tracking, account network analysis, and temporal coordination fingerprinting. Anthropic’s detection of the distillation attacks relied on exactly these signals [5]. Integrating coordination features into UBFS would address this second structural blind spot.

Adversarial-robust features. Experiment 9 shows that ASI06’s high detection rate is brittle—mimicry evasion drops LSTM from 0.976 to 0.731 AUC-ROC. Future work should develop features robust to nearest-neighbour mimicry, such as higher-order interaction terms or temporal signatures that cannot be individually replaced.

Adaptive window monitoring. Experiment 10 reveals that Isolation Forest saturates at 10 spans while LSTM benefits from full traces. A production system should use adaptive windows: short windows for fast threshold-based alerts (IF), long windows for temporal pattern detection (LSTM). The 10-span saturation point on ATBench suggests a principled default for MCP-style tool traces.

Transformer architectures. Self-attention [25] over behavioural sequences may capture longer-range dependencies than LSTM autoencoders and provide better interpretability through attention weight analysis. The LSTM’s robustness to decomposition evasion (Experiment 6, -0.007 vs. -0.051 to -0.064 for other models) suggests that sequence-aware architectures are more resistant to temporal evasion strategies—transformers may extend this advantage further.

VII. Conclusion

Seven findings, in order of surprise.

First, the transfer holds up—and generalises. An Isolation Forest trained on insider threat data is not degraded on agent traces (Experiment 2; the 0.019 change is within seed noise, and TRAIL’s $\sim 97\%$ prevalence makes its AUC-ROC an unstable estimand), and—on a balanced benchmark where the metric is trustworthy—transfer to MCP tool-calling traces measurably exceeds within-domain performance at 104.8% (Experiment 7, a gain $\sim 18\times$ the seed std). The UBFS bridge enables cross-domain anomaly detection across multiple agent benchmark boundaries, with the strongest evidence coming from the balanced-prevalence settings.

Second, one of our two “structural blind spots” was an artifact of our own generator. Experiment 3 identifies Tool Misuse (ASI02) as a blind spot at ~ 0.52 AUC-ROC, and Experiment 8’s semantic augmentation leaves it at ~ 0.56 . On 500 real ATBench trajectories it is the best-detected of six categories, at 0.82–0.95. The synthetic ranking is not merely miscalibrated but exactly inverted ($\rho = -1.000$ on the shared categories, consistently across all three models). The cause is visible in the generator: the ASI02 profile modifies only privilege features while preserving all behavioural structure, which makes it undetectable to a behavioural model by definition rather than by finding.

Third, the surviving blind spot is architectural, not behavioural. Distributed extraction (HYDRA) stays at ~ 0.54 because a per-entity detector scores each account’s marginal distribution while coordination lives in the joint—a limit of the monitoring architecture that no feature added to a per-account model can lift. It is corroborated by the Anthropic GTG-1002 disclosure, though not yet by a labelled benchmark. The distinction from ASI02 is the paper’s practical lesson: a blind spot argued from the structure of the monitor survived, and a blind spot measured from a perturbation we defined did not. Synthetic profiling generates hypotheses. It does not locate blind spots, and we would not have discovered that about

our own results without a real benchmark to check them against.

Fourth, evasion patterns transfer across domains. Task decomposition costs 5–6% detection power for IF and DC ($p=0.031$, Experiment 6), while LSTM shows near-invariance (-0.7% , $p=0.094$), mirroring the “boiling frog” pattern that drops insider threat detection from 69% to 3% in [4]. The Anthropic-disclosed GTG-1002 campaign used exactly this strategy. Distillation sensitivity analysis (Experiment 13) further shows that detection degrades monotonically with attack intensity for FOCUSED/BROAD/COT, but HYDRA sits at an optimal evasion intensity where attenuation paradoxically increases detection.

Fifth, high detection rates are brittle. Experiment 9 shows that mimicry evasion—copying a normal neighbour’s features while preserving the attack payload—degrades ASI06 detection by up to 25%. The categories that appear most detectable are the ones most vulnerable to informed adversaries: a detector that leans on a few loud features can have those features replaced.

Sixth, early signals suffice. Experiment 10 reveals that 10 spans capture the same detection signal as full traces for Isolation Forest on both TRAIL and ATBench. This mirrors the 7-day optimal window from insider detection [4]. Monitoring systems can use short observation windows without sacrificing accuracy, reducing both computational cost and detection latency.

Seventh, the framework generalises beyond OWASP. All five MITRE ATLAS techniques are strongly detectable through UBFS (≥ 0.93 AUC-ROC for IF/LSTM, Experiment 11). No new blind spots emerge. The ATLAS techniques—model replication, extraction, adversarial crafting—have strong volume and frequency signatures that UBFS captures by design. The only blind spot remains intent-based: attacks that use the right tools for the wrong reasons.

The deeper lesson: insider threat detection and AI agent monitoring are not analogous disciplines. They are the same discipline, applied to different principals. Building one from scratch while ignoring the other wastes decades of hard-won security knowledge—and risks rediscovering the same blind spots the hard way.

All code, data pipelines, and experimental results are available at <https://github.com/BipinRimal314/threat-to-governance-pipeline> to support reproducibility.

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